Confidence Intervals

Note: We have a RS $x_1 \dots x_n$ from a distribution/population with mean μ. (unknown) and variance σ^2 (known)

We want to "infer" about the distribution/population mean μ.

– statistical inference.

1) sample mean $\bar{x}$ is a "point estimate" of population/distribution mean μ.

2) A range of values in which μ is "likely" to fall is called a confidence interval.

<u>Example</u>: lifespan of vacuum tubes has an approximate normal distribution with mean μ hours and variance $\sigma^2 = 320^2$ hours^2.

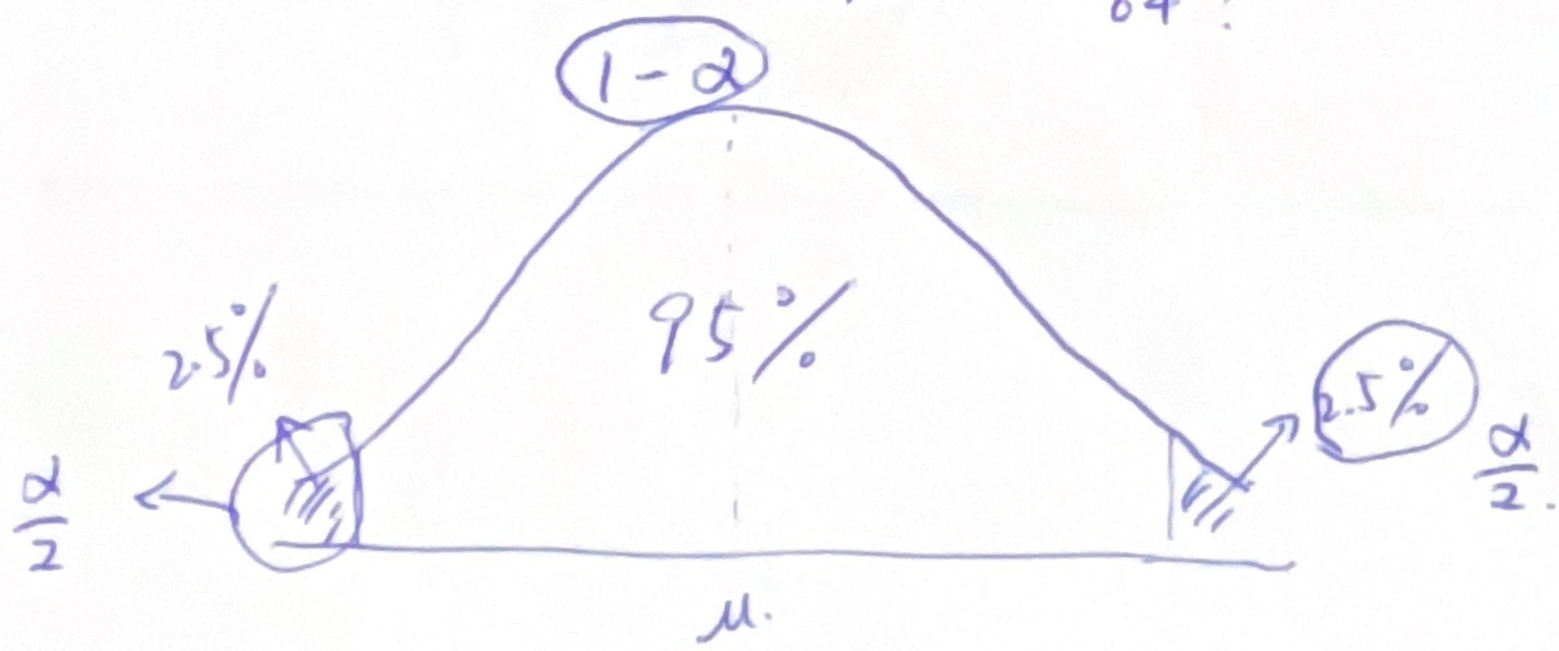
R. S. 64 tubes, give a point estimate.

$X_i \dot{\sim} N(\mu, \sigma^2 = 320^2)$ $i = 1, 2 \dots 64$.

$n = 64 \geq 30$

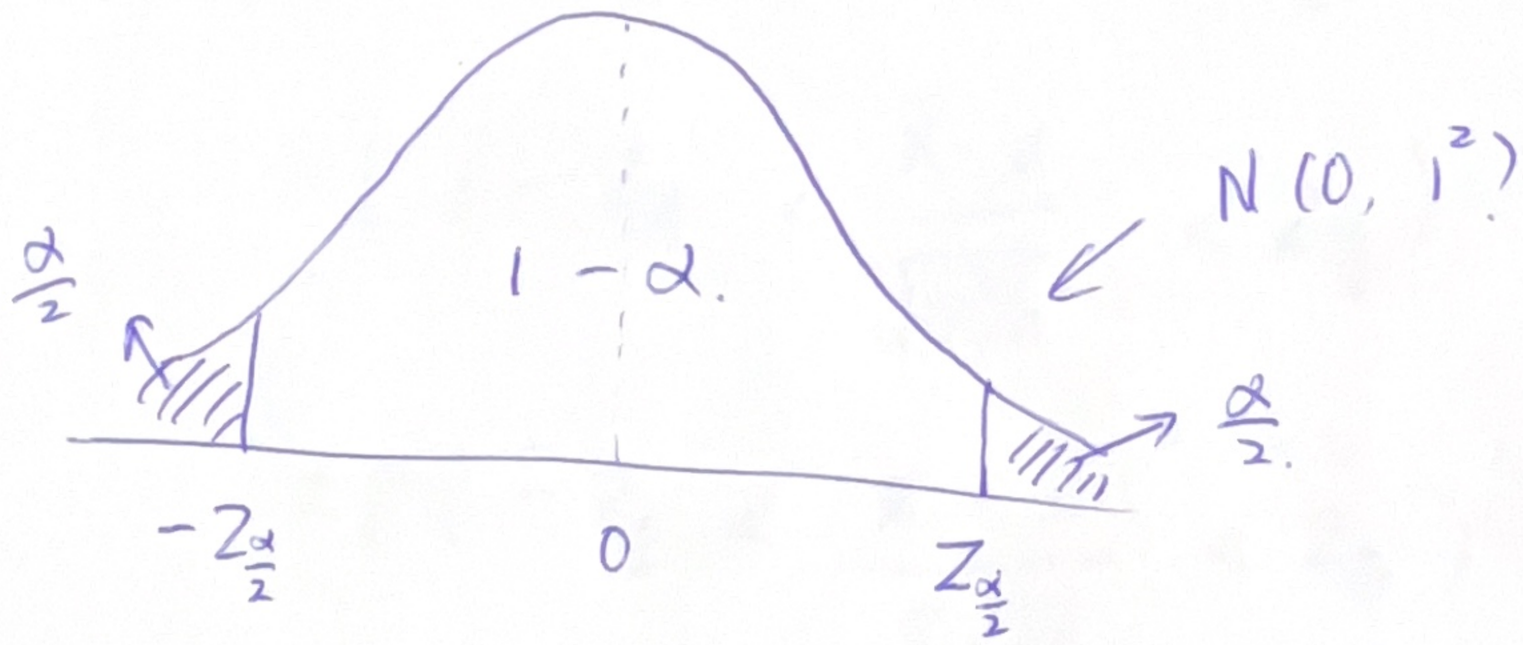
By CLT, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = 1478$ hours.

$\bar{X} \dot{\sim} N(\mu, \frac{\sigma^2}{n} = \frac{320^2}{64})$



Def: Let $Z_{\frac{\alpha}{2}}$ be the point on the number line such that

$$P(Z > Z_{\frac{\alpha}{2}}) = \frac{\alpha}{2}$$



$Z_{\frac{0.05}{2}} = Z_{0.025} = 1.96$ (95% CI)

$Z_{\frac{0.1}{2}} = Z_{0.05} = 1.645.$ (90% CI)

Theorem: CI for μ. (σ known)

Suppose $X_1 \ldots X_n$ is a RS from a normal distribution / population with mean μ (unknown) and variance σ^2 (known)

$$X_i \overset{iid}{\sim} N(\mu, \underbrace{\sigma^2}_{\text{known}}) \qquad i = 1, \ldots n.$$

then $(1-\alpha)100\%$ confidence interval for μ is

$$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}.$$

$$\bar{x} \sim N\left(\mu_{\bar{x}} = \mu,\ \sigma^2_{\bar{x}} = \frac{\sigma^2}{n}\right)$$

$$1-\alpha = P\left(-Z_{\frac{\alpha}{2}} < Z < Z_{\frac{\alpha}{2}}\right)$$

$$= P\left(-Z_{\frac{\alpha}{2}} < \frac{\bar{X}-\mu}{\sqrt{\sigma^2/n}} < Z_{\frac{\alpha}{2}}\right)$$

$$= P\left(-Z_{\frac{\alpha}{2}} < \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} < Z_{\frac{\alpha}{2}}\right)$$

$$= P\left(-Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} < \bar{x}-\mu < Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right)$$

$$= P\left(Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} > \mu-\bar{x} > -Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right)$$

$$= P\left(\bar{x}+Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} > \mu > \bar{x}-Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right)$$

A $(1-\alpha)100\%$ random interval for μ is

$\bar{x} \pm Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}$. [also proved to be the smallest interval]

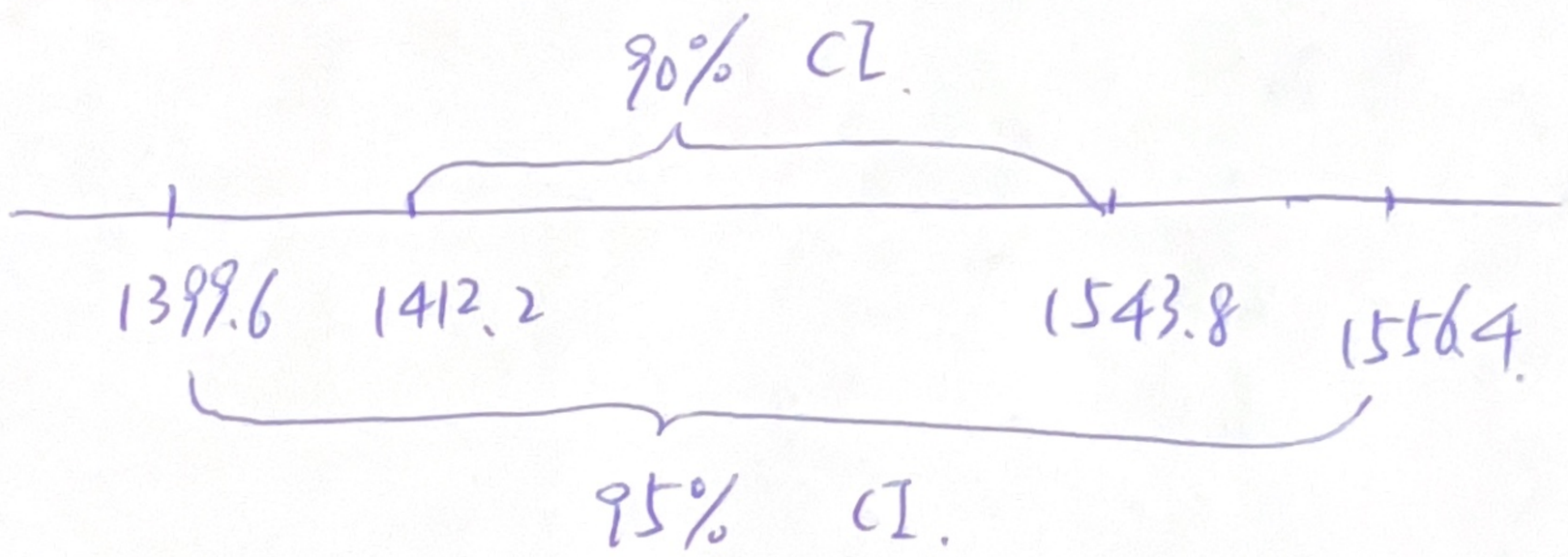
<u>Example:</u>

Find ~~CI~~ 95% CI for μ.

$$\bar{x} \pm Z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} = 1478 \pm 1.96 \times \frac{320}{\sqrt{64}} = (1399.6,\ 1556.4)$$

With 95% confidence, μ is within $(1399.6,\ 1556.4)$

Find 90% CI for μ.

$$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} = 1478 \pm 1.645 \times \frac{320}{\sqrt{64}} = (1412.2, 1543.8)$$

from 1.96 to 1.645